Question	Answers	Extra information	Mark	AO / Spec. Ref.
01.7 Mark with 01.8		max **3** marks for bar chart		AO2 4.1.3.2 RPA3
	correct scale **and** axis labelled (concentration (of salt solution) in mol/dm^3)	scale must take up at least 50% of grid	1	
	all points plotted correctly	allow a tolerance of ± ½ small square allow 3 or 4 correct plots for **1** mark	2	
	curved line of best fit	ignore line extended beyond 0.4 mol/dm^3 ignore line joined point to point with straight lines	1	

Question	Answers	Extra information	Mark	AO / Spec. Ref.
01.8 Mark with 01.7	correct answer from their line drawn on **Figure 1**	allow a tolerance of ± ½ small square ignore line joined point to point with straight lines if a line of best fit is drawn if no line of best fit is drawn, allow an answer in the range 0.31 – 0.33 (mol/dm^3)	1	AO2 4.1.3.2 RPA3

Question	Answers	Extra information	Mark	AO / Spec. Ref.
01.9		allow 'pieces' for potato throughout		AO2 4.1.3.2 RPA3
	water moves out of cells / potato		1	
	by osmosis	allow by diffusion through a partially / selectively / semi permeable membrane	1	
	(because) the solution in the cells / potato is less concentrated than outside **or** (because) the solution in the cells / potato is more dilute than outside	allow (because) the solution outside the cells / potato is more concentrated than inside allow (because) the solution outside the cells / potato is less dilute than inside allow correct references to water concentration / potential ignore reference to amount of water or salt do **not** accept water moves from an area of high (solute) concentration to an area of low (solute) concentration	1	

Total Question 1		**17**

Question 2

Question	Answers	Extra information	Mark	AO / Spec. Ref.
02.1	(cell) wall **or** (large / permanent) vacuole	ignore cellulose	1	AO3 4.1.1.1 4.1.1.2

Question	Answers	Extra information	Mark	AO / Spec. Ref.
02.2	rose black spot		1	AO1 4.3.1.4

Question	Answers	Extra information	Mark	AO / Spec. Ref.
02.3	$\frac{24 \times 60}{90}$ **or** $\frac{24}{1.5}$ 16	allow $\frac{1440}{90}$ do **not** accept if a unit is given	1 1	AO2 4.1.1.6 4.1.1.1

Question	Answers	Extra information	Mark	AO / Spec. Ref.
02.4	stomach		1	AO1 4.2.2.1

Question	Answers	Extra information	Mark	AO / Spec. Ref.
02.5	biuret reagent		1	AO1 4.2.2.1 RPA4

Question	Answers	Mark	AO / Spec. Ref.
02.6	**Level 3:** A judgement, strongly linked and logically supported by a sufficient range of correct reasons, is given.	5-6	AO3
	Level 2: Some logically linked reasons are given. There may also be a simple judgement.	3-4	AO2
	Level 1: Relevant points are made. They are not logically linked.	1-2	AO1
	No relevant content.	0	4.2.2.1 4.2.2.4 4.2.2.5 4.2.2.6

	Indicative content
	
	• meat-free burgers contain more fibre
	  • aids digestion **or** prevents constipation
	
	• meat burgers contain more protein
	  • for growth
	
	• meat burgers contain more fat
	  • can cause CHD **or** heart attack **or** narrowing of arteries
	  • may lead to needing a stent
	  • may lead to obesity
	  • obesity is a risk factor for (type 2) diabetes
	
	• meat burgers contain more cholesterol
	  • can cause narrowing of arteries **or** CHD **or** heart attack
	  • may lead to needing a stent
	  • may need to take statins
	
	• both burgers have similar amounts of carbohydrate
	  • good for providing energy
	
	• no information on vitamins / minerals provided for either burger
	
	• meat burgers require animals to be farmed
	  • increase in methane in atmosphere
	  • (methane) contributes to global warming
	
	• meat burgers require animals to be slaughtered
	  • ethical issues
	
	• some people won't eat meat-free burgers
	  • (because) some people don't like the idea of eating fungus
	  • (because) some people prefer the taste of meat
	
	For **Level 2**, comparisons and linked reasons using own knowledge are required.		

Total Question 2		**12**